Nikita Kirnosov

Senior Staff Software Engineer / Manager

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Skills

Technical Leadership, AI/ML Platforms Design, High Load Systems, Recommenders.

Languages C/C++, Python, Java, Go, SQL.

Experience

2024 Apr – Present **Senior Staff Software Engineer / Senior Engineering Manager**, *Google - Search*.
Created Search Discovery Foundations team composed of Updates Infrastructure, ML Platform, and AI Platform sub-teams.
- Grew the team by 20 engineers in 6 months.
- Landed real-time (15 minute fresh) transformer architecture recommendation models training and experimentation (+32% engagement).
- Built GenAI infrastructure in collaboration with Gemini and launched:
  - Topic updates generation: users receive meaningful updates on the followed queries (16% re-engagement increase).
  - *Samsung Now Bar*: users follow the games with real-time commentary generated from the UGC.
  - *Daily Listen*: daily AI-generated podcasts on the news personalized to your interests.
- Enhanced serving infrastructure to support live activities with industry-leading latencies.

2021 Apr – 2024 Apr **Staff Software Engineer / Engineering Manager**, *Google - Search*.
Led a Recommendations team powering notifications, promos, and suggestions, delivering >20% of inorganic Search growth in 2023. Established cross-org collaboration across 27 engineers from Google Research, Core Infrastructure, and Android — enabling 83 client team SWEs and PMs by modernizing user-content matching with transformer-based retrieval and LLM-powered generation.
- Reduced new update product idea-to-MVP time from a month to 2 hours and time to launch from a quarter to 2 weeks.
- Created a full cycle ML/AI ecosystem (indexing, generation, matching, training, serving).
  - Augmented heuristic-based content-user matching rules with ML based content retrieval, improving user engagement by 24%.
  - Created scalable architecture supporting various retrieval strategies (item2item, ScaNN, etc.), yielding 10% engagement increase in the first quarter.
- Introduced on-device ML to a stack, improving notification triggering quality by 7%.

2018 Apr – 2021 Apr **Senior Software Engineer**, *Google - Search*.
Envisioned and led the team implementing ML training and serving ecosystem for Search's News and Content Discovery.
- Designed and led the implementation of unified and standardized ML infrastructure (C++) for Discover, Notifications, Suggest and user modeling teams featuring:
  - zero training/serving skew near real time training data generation system, reducing the feature engineering work by 70%,
  - dynamic triggering rate optimization based on recent and long-term user activity pattern.
- Scaled an existing inference serving infrastructure to serve 24x more inferences using the same resources.
- Restructured Discover Search training process to support training on TPU to achieve 40x speedup (Python).
- Created user opt-out detection system proactively stopping bad experiments (C++).

2016 Apr – 2018 Apr **Software Engineer**, *Google - YouTube*.
Built storage abstraction microservices framework with no-downtime migration support (C++) and improved YT Public Data API performance, usability, and security (Python).
- Designed and implemented storage management framework allowing the user to create a full set of standard RESTful APIs for a defined resource in several hours regardless of the storage type (C++).
- Migrated video metadata migration between Vitess and Spanner with no downtime, data loss, or client latency increase (C++).

Sep 2015 – Oct 2015 **Fellow**, *Insight Data Science*.
Built *CityGravity.info*, a trip schedule optimization app to see the greatest number of attractions.

2010 Aug – 2016 Jan **Graduate Researcher**, *University of Arizona*, Tucson,AZ.
Developed Fortran code modeling the molecular system total wave function as an optimal combination of explicitly correlated Gaussian functions. Generalized the approach to be applied to exotic atoms and molecules and extended it to excited states. The results were published in over a dozen peer-reviewed papers and presented on multiple physics and supercomputing conferences.

Jan 2010 – May 2010 **Intern**, *Joint Institute of Nuclear Research*, Dubna,Russia.
Optimized muon detector signal processing code for the CMS experiment by vectorizing the core algorithm.

Education

2010–2015 **Ph.D.**, *University of Arizona*, Tucson, AZ – Physics.

2005–2010 **B.S. (Hons.)**, *Saratov State University*, Saratov, Russia – Physics.